all resulted in positive results. Most string theorists are convinced that the duality is true.

Following the same approach, one can study the strong coupling properties of another of the remaining string theories, say, the Type IIB string. As originally conjectured by Hull and Townsend and supported by the research of a number of physicists, something equally remarkable appears to occur. As the coupling constant of the Type IIB string gets larger and larger, the physical properties that we are still able to understand appear to match up exactly with that of the weakly coupled Type IIB string itself. In other words, the Type IIB string is self-dual.

Specifically, detailed analysis persuasively suggests that if the Type IIB coupling constant were larger than 1, and if we were to change its value to its reciprocal (whose value, therefore, is less than 1), the resulting theory is absolutely identical to the one we started with. Similar to what we found in trying to squeeze a circular dimension to a sub-Planck-scale length, if we try to increase the Type IIB coupling to a value larger than 1, the self-duality shows that the resulting theory is precisely equivalent to the Type IIB string with a coupling smaller than 1.

A Summary, So Far

Let’s see where we are. By the mid-1980s, physicists had constructed five different superstring theories. In the approximation scheme of perturbation theory, they all appear to be distinct. But this approximation method is valid only if the string coupling constant in a given string theory is less than 1. The expectation has been that physicists should be able to calculate the precise value of the string coupling constant in any given string theory, but the form of the approximate equations currently available makes this impossible. For this reason, physicists aim to study each of the five string theories for a range of possible values of their respective coupling constants, both less than and greater than 1—i.e., both weak and strong coupling. But traditional perturbative methods give no insight into the strong coupling characteristics of any of the string theories.

Recently, by making use of the power of supersymmetry, physicists have learned how to calculate some of the strong coupling properties